Numerical Analysis homework 1

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Due time: September 25, 2025

Abstract

My homework for solving nonlinear equations.

I. Bisection problems

I-a

The width of the interval at the nth step is $\frac{1}{2^{n-1}}$[1]

I-b

The supremum of the distance between the root r and the midpoint of the interval is $\frac{1}{2^n}$

II. Relative error of bisection problems

After n steps, the absolute error is bounded by:

$$|c_n - r| \leq \frac{b_0 - a_0}{2^{n+1}}$$

We require:

$$\left|\frac{c_n - r}{r}\right| \leq \epsilon$$

Since $r \geq a_0 > 0$:

$$\frac{b_0 - a_0}{2^{n+1}} \leq \epsilon a_0$$

$$2^{n+1} \geq \frac{b_0 - a_0}{\epsilon a_0}$$

$$n \geq \frac{\log\left(\frac{b_0 - a_0}{\epsilon a_0}\right)}{\log 2} - 1$$

$$n \geq \frac{\log(b_0 - a_0) - \log\epsilon - \log a_0}{\log 2} - 1$$

III. Perform four iterations of Newton's method

Given the polynomial equation:

$$p(x) = 4x^3 - 2x^2 + 3$$

with derivative:

$$p'(x) = 12x^2 - 4x$$

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Iteration	x_n	$p(x_n)$	$p'(x_n)$	x_{n+1}
0	-1.00000	-3.000	16.000	-0.81250
1	-0.81250	-0.464	11.170	-0.77100
2	-0.77100	-0.020	10.212	-0.76904
3	-0.76904	-0.002	10.168	-0.76884

Iteration 0: $\mathrm{p(x_0)} = 4(-1)^3 - 2(-1)^2 + 3 = -4 - 2 + 3 = -3$
$p'(x_0) = 12(-1)^2 - 4(-1) = 12 + 4 = 16$
$x_1 = -1 - \frac{-3}{16} = -1 + 0.1875 = -0.8125$

Iteration 1: $\mathrm{p(x_1)} = 4(-0.8125)^3 - 2(-0.8125)^2 + 3$
$= -2.144 - 1.320 + 3 = -0.464$
$p'(x_1) = 12(-0.8125)^2 - 4(-0.8125)$
$= 7.920 + 3.25 = 11.170$
$x_2 = -0.8125 - \frac{-0.464}{11.170}$
$= -0.8125 + 0.0415 = -0.7710$

Iteration 2: $\mathrm{p(x_2)} = 4(-0.7710)^3 - 2(-0.7710)^2 + 3$
$= -1.832 - 1.188 + 3 = -0.020$
$p'(x_2) = 12(-0.7710)^2 - 4(-0.7710)$
$= 7.128 + 3.084 = 10.212$
$x_3 = -0.7710 - \frac{-0.020}{10.212}$
$= -0.7710 + 0.00196 = -0.76904$

Iteration 3: $\mathrm{p(x_3)} = 4(-0.76904)^3 - 2(-0.76904)^2 + 3$
$= -1.820 - 1.182 + 3 = -0.002$
$p'(x_3) = 12(-0.76904)^2 - 4(-0.76904)$
$= 7.092 + 3.076 = 10.168$
$x_4 = -0.76904 - \frac{-0.002}{10.168}$
$= -0.76904 + 0.000197 = -0.76884$

IV. A variation of Newton's method

Let α be the true root, so the error at step n is:

$$e_n = x_n - \alpha$$

The error at step $n+1$ is:

$$e_{n+1} = x_{n+1} - \alpha = x_n - \frac{f(x_n)}{f'(x_0)} - \alpha$$

Using Taylor expansion of $f(x_n)$ around α:

$$f(x_n) = f(\alpha) + f'(\alpha)(x_n - \alpha) + \frac{f''(\xi_n)}{2}(x_n - \alpha)^2$$

Since $f(\alpha) = 0$:

$$f(x_n) = f'(\alpha)e_n + \frac{f''(\xi_n)}{2}e_n^2$$

$$e_{n+1} = e_n - \frac{f'(\alpha)e_n + \frac{f''(\xi_n)}{2}e_n^2}{f'(x_0)}$$

$$e_{n+1} = \left(1 - \frac{f'(\alpha)}{f'(x_0)}\right)e_n - \frac{f''(\xi_n)}{2f'(x_0)}e_n^2$$

For sufficiently small errors, the second-order term may be ignored as it becomes negligible.

$$s = 1, \quad C = 1 - \frac{f'(\alpha)}{f'(x_0)}$$

V.Convergence Analysis

Although $g'(0) = 1$,but for $x \neq 0$:

$$|g'(x)| = \left|\frac{1}{1+x^2}\right| < 1 \quad and \quad |\arctan x| < |x|$$

Therefore, the iteration is a contraction mapping.It converges.

VI.Convergence Analysis

Consider the function:

$$f(x) = \frac{1}{p+x}$$

Its derivative is:

$$f'(x) = -\frac{1}{(p+x)^2}$$

For $x > 0$ and $p > 1$, we have:

$$|f'(x)| = \frac{1}{(p+x)^2} < \frac{1}{p^2} < 1$$

Thus, f is a contraction mapping on $(0, \infty)$, guaranteeing convergence.

$$x = \frac{1}{p+x}$$

This can be rewritten as:

$$x(p+x) = 1$$
$$x^2 + px - 1 = 0$$

The quadratic equation has solutions:

$$x = \frac{-p \pm \sqrt{p^2+4}}{2}$$

Since $p > 1$ and $x > 0$ (as all terms are positive), we take the positive root:

$$x = \frac{-p + \sqrt{p^2+4}}{2}$$

And it's the answer.

VII.Another Analysis for II

Case: $a_0 < 0 < b_0$ When the root r is close to or equal to zero, the relative error measure:

$$RelativeError = \frac{|x_n - r|}{|r|}$$

becomes problematic because the denominator $|r|$ approaches zero.

So when r is near zero, we should use absolute error instead:

$$|x_n - r| \leq \epsilon$$

The absolute error bound for bisection method is:

$$|x_n - r| \leq \frac{b_0 - a_0}{2^{n+1}}$$

To guarantee absolute error $\leq \epsilon$:

$$\frac{b_0 - a_0}{2^{n+1}} \leq \epsilon$$
$$2^{n+1} \geq \frac{b_0 - a_0}{\epsilon}$$
$$n + 1 \geq \log_2\left(\frac{b_0 - a_0}{\epsilon}\right)$$
$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} - 1$$

VIII.a Root of Multiplicity k in Newton's Method

Let r be a root of multiplicity $k > 1$ of $f(x)$. Then $f(x)$ can be written as:

$$f(x) = (x-r)^k g(x), \quad where g(r) \neq 0.$$

The derivative is:

$$f'(x) = k(x-r)^{k-1}g(x) + (x-r)^k g'(x) = (x-r)^{k-1}[kg(x) + (x-r)g'(x)].$$

The iteration is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Let $e_n = x_n - r$ be the error at step n. Then:

$$e_{n+1} = e_n - \frac{f(x_n)}{f'(x_n)} = e_n - \frac{(x_n-r)^k g(x_n)}{(x_n-r)^{k-1}[kg(x_n) + (x_n-r)g'(x_n)]} = e_n - \frac{e_n^k g(x_n)}{e_n^{k-1}[kg(x_n) + e_n g'(x_n)]} = e_n - \frac{e_n g(x_n)}{kg(x_n) + e_n g'(x_n)}.$$

So:

$$e_{n+1} = e_n \left[\frac{kg(x_n) + e_n g'(x_n) - g(x_n)}{kg(x_n) + e_n g'(x_n)}\right] = e_n \left[\frac{(k-1)g(x_n) + e_n g'(x_n)}{kg(x_n) + e_n g'(x_n)}\right].$$

Limit as $e_n \to 0$ As $x_n \to r$ (so $e_n \to 0$) and since $g(x)$ is continuous with $g(r) \neq 0$:

$$\lim_{e_n \to 0} \frac{(k-1)g(x_n) + e_n g'(x_n)}{kg(x_n) + e_n g'(x_n)} = \frac{(k-1)g(r)}{kg(r)} = 1 - \frac{1}{k}.$$

Therefore, for small e_n:

$$e_{n+1} \approx \left(1 - \frac{1}{k}\right) e_n.$$

Part 1: Detecting Multiple Zeros

A multiple zero can be detected by observing the convergence behavior:

- For a simple root ($k = 1$), Newton's method exhibits quadratic convergence:

  $$|e_{n+1}| \approx C|e_n|^2$$

  where $e_n = x_n - r$.

- For a multiple root ($k > 1$), standard Newton's method shows linear convergence:

  $$|e_{n+1}| \approx \left(1 - \frac{1}{k}\right)|e_n|$$

- Diagnostic approach: Compute the ratio $\frac{|e_{n+1}|}{|e_n|}$. If it approaches a constant $\in (0,1)$ rather than 0, this suggests a multiple root.

Part 2: Proof of Quadratic Convergence for Modified Newton's Method

$$x_{n+1} = x_n - k\frac{f(x_n)}{f'(x_n)} = x_n - k\frac{e_n^k g(x_n)}{e_n^{k-1}[kg(x_n) + e_n g'(x_n)]}$$

$$e_{n+1} = e_n - k\frac{e_n g(x_n)}{kg(x_n) + e_n g'(x_n)} = e_n \left[1 - \frac{kg(x_n)}{kg(x_n) + e_n g'(x_n)}\right]$$

$$e_{n+1} = e_n \left[\frac{e_n g'(x_n)}{kg(x_n) + e_n g'(x_n)}\right] \approx \frac{g'(r)}{kg(r)} e_n^2 \quad (quadratic convergence)$$

Thus, the modified Newton's method restores quadratic convergence for multiple roots.